MIL-C-60839(MU)
10 October 1968

SUPERSEDING
JAN-C-746 (In Part)
30 March 1949
(See 6.2)

MILITARY SPECIFICATION
CARTRIDGE, GRENADE: CARBINE, CALIBER .30, M6

1. SCOPE

1.1 This specification covers Cartridge, Grenade: Carbine, Caliber .30, M6 intended for propelling grenades from a launcher attached to carbine weapons.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-636	- Visual Inspection Standards for Small Arms Ammunition through Caliber .50
MIL-STD-644	- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition

DRAWINGS

U.S. Army Munitions Command

B6181079	- Cartridge, Grenade, Carbine, Caliber .30, M6
C7643674	- Classification of Cartridge Case Defects
IEL-6181079	- Inspection Equipment List for Cartridge, Grenade, Carbine, Caliber .30, M6

FSC 1330

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4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 200 cartridges. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Lot identification.- Each lot of ammunition shall be identified as to type, caliber, and model as well as with a lot number and the supplier's identification as assigned by the procuring activity. Each lot shall be further identified by a Federal Stock Number assigned by the procuring activity.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The Acceptable Quality Level (AQL) for the major class shall be 0.25 percent and the AQL for the minor class shall be 1.50 percent. All non-conforming cartridges shall be rejected.

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4.3.2.1 (Cont'd)

TABLE I (Cont'd)

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Gaging				
38	Cartridge profile failure (requiring more than 20 lbs. dead weight to insert in profile and alignment gage)		X		
39	Diameter of extractor groove			X	
40	Diameter of head		X		
41	Thickness of head, max.			X	
42	Thickness of head, min.		X		
43	Depth of primer		X		
44	Length to shoulder		X		
	Weighing				
45	Weight, min.	X			

1/ Refer to MIL-STD-636 for visual defect standards for defects 1 through 32. While no photographs of grenade cartridges are included in this standard, the cartridge case defects illustrated in the Caliber .30 Carbine section can be used. In the event of conflict between Table I of this specification and MIL-STD-636 as to defect classification, the classification specified in Table I shall apply.

4.3.3 Tests.- The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.

4.3.3.1 Test samples.- The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual, dimensional and weight requirements shall be used in the ballistic tests, and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges shall be thoroughly mixed before being divided into samples for the various tests.

TABLE II

Test	Number of Cartridges	Requirement Paragraph
Velocity 1/	25	3.2
Function and Casualty 2/	--	3.3

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4.3.3.2 (Cont'd)

TABLE III (Cont'd)

Defects	Acceptance Number First Sample	Acceptance Number Cumulative (First & Second Sample)
3. Case casualties 1/		
a. Longitudinal split 2/		
(1) Body (J)	0	2
(2) Body (K)	0	1
(3) To head (L)	0	1
(4) Through head (M)	0	1
b. Circumferential rupture 2/		
(1) Partial, body (J)	0	2
(2) Partial, body (K)	0	1
(3) Partial, head (L)	0	1
(4) Complete	0	1
4. Failure to extract	0	1

1/ No penalty shall be charged against a fired case with mouth partially or completely burned. If, however, such fired cartridge case shows the occurrence of some other firing casualty, exclusive of failure of rosebud crimp to open, such other casualty shall be penalized in accordance with Table III.

2/ For location of defects indicated by letters in parentheses, see Drawing C7643674.

4.3.4 Packaging, packing and marking inspection.- During or immediately prior to the packaging operation, 100% examination of the cartridges shall be performed to ascertain the the cartridge type conforms to the drawing. Occurrence of a dummy, blank or any bullet type cartridge shall be classed as a critical defect. Occurrence of any incorrect type other than those listed shall be classed as a major defect. All non-conforming cartridges shall be rejected. Inspection for packaging, packing and marking shall be in accordance with Standard MIL-STD-644 as applicable to the drawing.

4.3.5 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-6181079.

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6.1.5 Requirement for contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection System Requirements.

6.2 Supersession data.- This specification together with MIL-C-746A(MU) Cartridge, Grenade, Rifle, Caliber .30, M3 supersedes JAN-C-746 dated 30 March 1949. Cartridge, Grenade, Auxiliary, M7 has been reclassified to obsolete status.

Custodian:

Army - MU

Preparing activity:

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